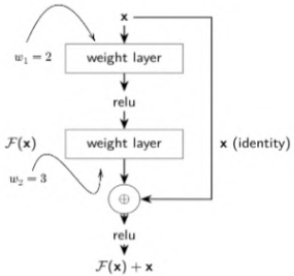
Question 1 / 10 Flag for review

1 mark

Common Data for Question 1-2

Consider the below network architecture which uses the skip connection.



If the input to the network is x=−1 then what will be the output of the network before skip connection? There are no bias attached to the nextwork.

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Your answer

Enter your answer

Only numeric input is accepted for this question.

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With the same input (i.e. x=−1), what will be the output after skip connection?

Your answer

Enter your answer

Only numeric input is accepted for this question.

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Question 3 / 10 Flag for review • 1 mark

In a semantic segmentation model, what is the typical role of the 'decoder'?

Options

A. gradually up-sample the low-resolution feature maps to produce a full-resolution segmentation map.

B. To apply data augmentation to the input images before they enter the network.

C. To extract high-level semantic features from the input image.

D. To calculate the final loss function and update the model's weights.

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Question 4 / 10 Flag for review • 1 mark

Consider the code for setting up a model for transfer learning:

```
model = torchvision.models.efficientnet_b0(weights="DEFAULT")
for p in model.parameters():
    p.requires_grad = False

model.classifier[1] = torch.nn.Linear(1280, 3)
optimizer = torch.optim.Adam(model.parameters(), lr=3e-3)
```

When optimizer.step() is called during training, which parameters will be updated?

Options

A. None, because the first loop set requires_grad = False on all original parameters.

B. Only the weights and bias of the new torch.nn.Linear(1280, 3) layer.

C. All parameters in model.classifier, including the Dropout layer.

D. All model parameters, because the optimizer was instantiated with model.parameters().

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Question 5 / 10 Flag for review • 1 mark

Which of the following transformation is wrong for the test dataset? Assume the following imports.

```
1 import albumentations as A
```

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Options

A. Resize(512,512)

B. Normalize(mean=(0,0,0), std=(1,1,1))

C. Rotate(p=0.1)

D. ToTensorV2()

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Question 6 / 10 Flag for review • 1 mark

What is the expected directory format for using datasets.ImageFolder?

Options

A. train/class_instances

B. all_classes/train/instances

C. train/<class_name>/<images>

D. images/train/class_name

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Question 7 / 10 Flag for review • 1 mark

For a PIL input image transformation consider the below code snippet. Assume that image_mean and image_std are properly well defined.

```
import torchvision.transforms as transforms
def transform_data():
 train_transform = transforms.Compose([
    transforms.Resize((100,100)),
    transforms.RandomHorizontalFlip(p=0.1),
    transforms.RandomRotation(degrees=5),
    transforms.ColorJitter(
       brightness = 0.2,
       saturation = 0.2
    ),
    transforms.Normalize(mean=image_mean, std=image_std),
    transforms.ToTensor()
 ])
 return train_transform
```

What is wrong with the above code?

Options

A. `transforms.Resize must always come after ToTensor()`

B. `transforms.RandomRotation cannot be combined with transforms.ColorJitter.`

C. Nothing is wrong, the code will work fine as it is.

D. `transforms.Normalize is applied before ToTensor().`

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Question 8 / 10 Flag for review • 1 mark

The outputs of the parallel branches in a GoogLeNet Inception module are combined by:

Options

A. Element-wise addition of all feature maps.

B. Concatenation of the feature maps along the channel dimension.

C. A 1x1 convolution that merges all feature maps.

D. A global average pooling operation.

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Question 9 / 10 Flag for review • 1 mark

Consider this attempt at a residual connection:

```
import torch.nn as nn
import torch

# A block that halves spatial dimensions and doubles channels
block = nn.Sequential(
   nn.Conv2d(64, 128, kernel_size=3, stride=2, padding=1), nn.ReLU(),
   nn.Conv2d(128, 128, kernel_size=3, padding=1), nn.ReLU()
)
x = torch.randn(8, 64, 56, 56)
output = block(x) + x
```

What is the result of executing this code?

Options

A. It executes successfully, as PyTorch will broadcast x to match the shape of block(x).

B. It fails with a runtime error because the spatial dimensions of x (56x56) and block(x) (28x28) are incompatible for addition.

C. It fails with a runtime error because the channel dimensions of x (64) and block(x) (128) are incompatible for addition.

D. It fails with a runtime error because both the spatial and channel dimensions are incompatible for element-wise addition.

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Question 10 / 10 Flag for review

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What is the output of the final print statement of the following code?

```
import torch
import torch.nn as nn
import torchvision.models as models

model = models.vgg16(pretrained=True)

# Freeze feature extractor
for param in model.parameters():
    param.requires_grad = False

# Adding a linear layer for fine tuining (including bias)
model.classifier[6] = nn.Linear(4096, 5, bias = False)

# Count trainable parameters
trainable_params = sum(p.numel() for p in model.parameters() if p.requires_grad)
print(trainable_params)
```

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Your answer

Enter your answer

Only numeric input is accepted for this question.

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